1️⃣ Method Overriding in C#

What is Method Overriding?

Method Overriding occurs when a **derived class provides a new implementation for a method that is already defined in the base class**.

The method in the base class must be declared using **virtual**, and the derived class must use **override**.

Definition

Method overriding allows a derived class to change the behavior of a base class method.

Syntax

```
class BaseClass
{
    public virtual void MethodName()
    {
    }
}

class DerivedClass : BaseClass
{
    public override void MethodName()
    {
    }
}
```

Example of Method Overriding

Base Class

```
using System;


class Animal
{
    public virtual void MakeSound()
    {
        Console.WriteLine("Animal makes sound");
    }
}
```

Derived Class

```
class Dog : Animal
{
    public override void MakeSound()
    {
        Console.WriteLine("Dog barks");
    }
}
```

Main Program

```
class Program
{
    static void Main()
    {
        Animal a = new Dog();
        a.MakeSound();
    }
```

```
}
```

Output

Dog barks

Notice something important:

Animal a = new Dog();

Even though the **reference type is Animal**, the **Dog method executes**.

This is called **Runtime Polymorphism**.

Key Characteristics of Method Overriding

Feature	Description
Uses	virtual and override
Binding	Runtime Binding
Polymorphism	Yes
Same method signature	Required
Same return type	Required

Real-world Example

Healthcare system example.

```
class HospitalStaff
{
    public virtual void Work()
    {
        Console.WriteLine("Staff working");
    }
}
```

```
class Doctor : HospitalStaff
{
    public override void Work()
    {
        Console.WriteLine("Doctor treating patient");
    }
}
```

2️⃣ Method Hiding in C#

What is Method Hiding?

Method hiding occurs when the **derived class defines a method with the same name as the base class method**, but it **does not override it**.

Instead, it **hides the base class method** using the **new keyword**.

Definition

Method hiding hides the base class method rather than replacing it.

Syntax

```
class BaseClass
{
    public void Show()
    {
    }
}

class DerivedClass : BaseClass
```

```
{
    public new void Show()
    {
    }
}
```

Example of Method Hiding

Base Class

```
using System;


class Animal
{
    public void MakeSound()
    {
        Console.WriteLine("Animal makes sound");
    }
}
```

Derived Class

```
class Dog : Animal
{
    public new void MakeSound()
    {
        Console.WriteLine("Dog barks");
    }
}
```

Main Program

```
class Program
{
    static void Main()
    {
        Animal a = new Dog();
        Dog d = new Dog();

        a.MakeSound();
        d.MakeSound();
    }
}
```

Output

Animal makes sound

Dog barks

Explanation:

Animal a = new Dog();

The **reference type decides which method executes**, not the object type.

So **base class method runs**.

Key Characteristics of Method Hiding

Feature	**Description**
Uses	new keyword
Binding	Compile-time binding
Polymorphism	No
Base method replaced	No

Feature	Description
Base method hidden	Yes

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3 Step-by-Step Comparison

Feature	Method Overriding	Method Hiding
Keyword	override	new
Base Method	virtual required	Not required
Binding	Runtime	Compile-time
Polymorphism	Yes	No
Behavior	Replaces base method	Hides base method
Reference Type Effect	Object decides	Reference decides

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4 Visual Understanding

Method Overriding

Animal a = new Dog();

a.MakeSound();

Runtime checks object type → **Dog method runs**

Method Hiding

Animal a = new Dog();

a.MakeSound();

Compile-time checks reference type → **Animal method runs**

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5 Example Showing Both Together

```
using System;

class Parent
{
    public virtual void Display()
    {
        Console.WriteLine("Parent Method");
    }

    public void Show()
    {
        Console.WriteLine("Parent Show");
    }
}

class Child : Parent
{
    public override void Display()
    {
        Console.WriteLine("Child Display");
    }

    public new void Show()
    {
        Console.WriteLine("Child Show");
    }
```

```
}

class Program
{
    static void Main()
    {
        Parent p = new Child();

        p.Display();
        p.Show();
    }
}
```

Output

Child Display

Parent Show

Explanation:

Method	**Type**	**Output**
Display	Overriding	Child Display
Show	Hiding	Parent Show

Method Overriding

Method overriding allows a derived class to provide a specific implementation of a base class method using virtual and override.

Method Hiding

Method hiding allows a derived class to hide a base class method using the new keyword.

7️⃣ When to Use

Use Method Overriding when

✓ You want **polymorphic behavior**
✓ Base method behavior should change

Example:

- Payment processing
- Shape area calculation
- Employee salary calculation

Use Method Hiding when

✓ You want **independent methods**
✓ No polymorphism required